

Q3A: Derive Hamilton's equations of motion for a particle in a conservative force field using Hamilton's principle.

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1. Setup and Definitions

Consider a particle of mass m moving in three dimensions under a conservative force $\mathbf{F} = -\nabla V(\mathbf{q})$, where V is the potential energy and $\mathbf{q} = (q_1, q_2, q_3)$ are generalised coordinates.

Definition (Lagrangian). The **Lagrangian** is

$$L(\mathbf{q}, \dot{\mathbf{q}}) = T - V = \frac{1}{2}m \sum_{i=1}^3 \dot{q}_i^2 - V(\mathbf{q}),$$

where T is the kinetic energy and dots denote time derivatives.

Definition (Generalised Momentum). The **generalised (conjugate) momentum** corresponding to q_i is

$$p_i = \frac{\partial L}{\partial \dot{q}_i} = m\dot{q}_i.$$

The pair (q_i, p_i) forms a set of *canonical variables*.

Definition (Hamiltonian). The **Hamiltonian** is the Legendre transform of L with respect to $\dot{\mathbf{q}}$:

$$H(\mathbf{q}, \mathbf{p}) = \sum_{i=1}^3 p_i \dot{q}_i - L(\mathbf{q}, \dot{\mathbf{q}}),$$

where $\dot{q}_i$ is expressed in terms of p_i via $\dot{q}_i = p_i/m$. For our particle,

$$H = \sum_i p_i \cdot \frac{p_i}{m} - \left(\frac{1}{2m} \sum_i p_i^2 - V \right) = \frac{|\mathbf{p}|^2}{2m} + V(\mathbf{q}) = T + V.$$

The Hamiltonian equals the total mechanical energy.

2. Hamilton's Principle (Principle of Stationary Action)

Definition (Action Functional). The **action** along a path $\mathbf{q}(t)$ between fixed endpoints $\mathbf{q}(t_1)$ and $\mathbf{q}(t_2)$ is

$$S[\mathbf{q}] = \int_{t_1}^{t_2} L(\mathbf{q}, \dot{\mathbf{q}}) dt.$$

Hamilton's Principle states: the physical trajectory of the system is the path $\mathbf{q}(t)$ for which the action S is *stationary*, i.e. $\delta S = 0$, subject to fixed endpoints $\delta \mathbf{q}(t_1) = \delta \mathbf{q}(t_2) = 0$.

3. From Lagrangian to Hamiltonian Formulation

We rewrite the action in terms of the canonical variables $(\mathbf{q}, \mathbf{p})$ by substituting $L = \sum_i p_i \dot{q}_i - H$:

$$S = \int_{t_1}^{t_2} \left(\sum_{i=1}^3 p_i \dot{q}_i - H(\mathbf{q}, \mathbf{p}) \right) dt.$$

We now treat $\mathbf{q}(t)$ and $\mathbf{p}(t)$ as *independent* variations.

4. Derivation of Hamilton's Equations

Theorem 1 (Hamilton's Equations of Motion). *The stationarity condition $\delta S = 0$ under independent variations δq_i and δp_i (with $\delta q_i(t_1) = \delta q_i(t_2) = 0$) yields*

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}, \quad i = 1, 2, 3.$$

Proof. Compute the first variation of S :

$$\delta S = \int_{t_1}^{t_2} \sum_i \left(\delta p_i \dot{q}_i + p_i \delta \dot{q}_i - \frac{\partial H}{\partial q_i} \delta q_i - \frac{\partial H}{\partial p_i} \delta p_i \right) dt.$$

Step 1 — Integrate the $p_i \delta \dot{q}_i$ term by parts.

$$\int_{t_1}^{t_2} p_i \delta \dot{q}_i dt = \left[p_i \delta q_i \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \dot{p}_i \delta q_i dt.$$

The boundary term vanishes since $\delta q_i(t_1) = \delta q_i(t_2) = 0$.

Step 2 — Collect terms.

Substituting back:

$$\delta S = \int_{t_1}^{t_2} \sum_i \left[\left(\dot{q}_i - \frac{\partial H}{\partial p_i} \right) \delta p_i - \left(\dot{p}_i + \frac{\partial H}{\partial q_i} \right) \delta q_i \right] dt = 0.$$

Step 3 — Apply the Fundamental Lemma of Calculus of Variations.

Since δq_i and δp_i are *independent and arbitrary*, each coefficient must vanish identically:

$$\begin{aligned} \delta p_i \text{ arbitrary: } \quad \dot{q}_i - \frac{\partial H}{\partial p_i} = 0 &\implies \boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}}, \\ \delta q_i \text{ arbitrary: } \quad \dot{p}_i + \frac{\partial H}{\partial q_i} = 0 &\implies \boxed{\dot{p}_i = -\frac{\partial H}{\partial q_i}}. \end{aligned}$$

These are Hamilton's canonical equations of motion. □

5. Explicit Form for a Particle in a Conservative Field

For $H = \frac{|\mathbf{p}|^2}{2m} + V(\mathbf{q})$, the partial derivatives are

$$\frac{\partial H}{\partial p_i} = \frac{p_i}{m}, \quad \frac{\partial H}{\partial q_i} = \frac{\partial V}{\partial q_i}.$$

Hamilton's equations therefore give:

$$\dot{q}_i = \frac{p_i}{m} \iff p_i = m\dot{q}_i \quad (\text{momentum definition}),$$

$$\dot{p}_i = -\frac{\partial V}{\partial q_i} \iff m\ddot{q}_i = -\frac{\partial V}{\partial q_i} = F_i \quad (\text{Newton's second law}).$$

The two first-order equations together reproduce Newton's second law $m\ddot{\mathbf{q}} = -\nabla V$, confirming the derivation is consistent.

6. Conservation of Energy

Using Hamilton's equations, the total time derivative of H is

$$\frac{dH}{dt} = \sum_i \left(\frac{\partial H}{\partial q_i} \dot{q}_i + \frac{\partial H}{\partial p_i} \dot{p}_i \right) + \frac{\partial H}{\partial t} = \sum_i \left(\frac{\partial H}{\partial q_i} \cdot \frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i} \cdot \frac{\partial H}{\partial q_i} \right) + \frac{\partial H}{\partial t}.$$

The sum cancels identically, leaving

$$\frac{dH}{dt} = \frac{\partial H}{\partial t}.$$

For a conservative field $V = V(\mathbf{q})$ (no explicit time dependence), $\partial H/\partial t = 0$, so

$$\frac{dH}{dt} = 0 \implies H = E = \text{constant}.$$

The Hamiltonian (total energy) is conserved along every physical trajectory.

7. Summary

Starting from Hamilton's principle $\delta S = 0$ with action $S = \int_{t_1}^{t_2} (\sum_i p_i \dot{q}_i - H) dt$, independent variation of $\mathbf{q}$ and $\mathbf{p}$ and integration by parts yields the **canonical equations**:

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}.$$

For a particle in a conservative field with $H = |\mathbf{p}|^2/2m + V(\mathbf{q})$, these reduce to Newton's second law. The formulation replaces one second-order ODE per coordinate with two first-order ODEs in the canonical pair (q_i, p_i) , and reveals the deep symmetry between coordinates and momenta that underlies classical mechanics.